

ADVANCED FINANCIAL ANALYSIS, PROBABILISTIC MODELING, AND STATISTICAL VALIDATION OF BIG TECH STOCKS

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TABLE OF CONTENTS

1. INTRODUCTION AND PROJECT SCOPE
2. RELATIVE PERFORMANCE AND BASE-100 NORMALIZATION
3. THE EVOLUTION OF QUANTITATIVE PERFORMANCE METRICS AND NORMALIZATION
4. STOCHASTIC FORECASTING VIA MONTECARLO SIMULATION
5. INDIVIDUAL ANALYSIS OF THE MONTECARLO SIMULATIONS
6. ANALYTICAL FRAMEWORK: THE STANDARD ROLE OF HISTORICAL NORMALIZATION IN ASSET MANAGEMENT
7. ADVANCED STATISTICAL SUMMARY AND DISTRIBUTION FINDINGS
8. ECONOMETRIC HYPOTHESIS FRAMEWORK
9. STRATEGIC PURPOSE AND QUANTITATIVE CONCLUSIONS
10. PERSONAL MOTIVATION AND ACKNOWLEDGMENTS

1. INTRODUCTION AND PROJECT SCOPE

This project is a personal initiative designed to bridge the gap between financial theory, quantitative mathematics, and practical programming. The primary objective is to build a foundational understanding of portfolio management and asset allocation. To achieve this, the project introduces a quantitative framework designed to assess the financial performance, potential future trajectories, and underlying return distributions of three dominant mega-cap technology equities: Apple, Microsoft, and Google. By cross-examining historical data spanning a three-year timeline against the broader market index, the Standard & Poor's 500, this research develops a pipeline encompassing investment benchmarking, stochastic simulations, empirical density exploration, and formal econometric hypothesis validation.

Although this framework is introductory and relatively basic, it remains highly functional and useful. It serves as an excellent case study to present to a potential client when advising them on an investment in one of these three companies. By shifting the conversation from speculative guesswork to data-driven analysis, it provides a clear, transparent, and mathematical foundation to help clients understand risk, performance, and future probability boundaries before they deploy real capital.

2. RELATIVE PERFORMANCE AND BASE-100 NORMALIZATION

The Base-100 normalization principle is a foundational quantitative tool in asset management, performance benchmarking, and modern portfolio theory. In financial markets, conducting a direct comparison of nominal equity prices introduces severe structural scaling biases. An asset trading at a nominal price of 400 dollars per share is not inherently superior, more valuable, or structurally more volatile than an asset trading at 10 dollars per share.

A stock's nominal price is largely an arbitrary figure dictated by historical corporate actions, such as share issuance volumes and stock splits. Consequently, evaluating the absolute price charts of different companies simultaneously lacks economic meaning. From an investment perspective, the absolute cost per individual unit is irrelevant; what matters fundamentally is the rate of return on the total capital deployed.

To eliminate nominal scaling biases and visualize a mathematically equitable race from an identical starting gate, raw closing price series must be standardized. Normalization achieves this by transforming a time-series of absolute historical prices into a synchronized series of index values, forcing the initial baseline value to a fixed constant, typically 100.

The algebraic transformation for any given timestamp t is mathematically defined as follows:

Normalized Price at time t equals the closing price at time t divided by the initial closing price at time zero, with the entire result multiplied by 100.

Where P_t represents the asset's closing valuation at timestamp t , and P_0 marks the absolute initial price vector on the very first day of the historical lookback period.

Practical Utility and Analytical Value

By applying this mathematical constraint, the original currency dimension is stripped away, turning the data into a pure metric of cumulative percentage growth. This methodology provides several distinct advantages for both quantitative analysts and clients:

True Performance Comparison: It forces all selected equities to begin at an identical point of origin. This isolates pure capital appreciation, enabling an objective, side-by-side comparison of how efficiently different companies grow capital over time.

Instant Monetary Translation: It translates abstract percentage returns into a highly tangible scenario for investors. Because the baseline is 100, the vertical axis of the normalized chart maps directly to the real-time value of a hypothetical initial investment of 100 dollars. If the index stands at 145, the client instantly sees a 45 percent net profit.

Effective Benchmarking: It allows individual equities or custom portfolios to be seamlessly plotted against broad market benchmarks, such as the S&P 500 index, regardless of how massive the differences in their nominal trading prices might be.

The following sections provide a rigorous step-by-step mathematical induction proof, an intuitive practical example, and a review of the institutional applications of this model.

Mathematical Induction Proof

To formalize that the Base-100 index structure accurately and consistently preserves the exact properties of relative asset returns over time, we can prove through mathematical induction that the index value at any arbitrary timestamp t is exactly equivalent to applying successive historical daily returns to the initial baseline.

Preliminary Definitions and Algebraic Setup

First, we define the simple daily return of the asset during period t as the relative price change from the previous day:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1 \implies \frac{P_t}{P_{t-1}} = 1 + R_t$$

Where P_t is the closing price at day t , and $P_{(t-1)}$ is the closing price of the preceding session. The term $(1 + R_t)$ represents the daily growth factor.

Second, the standard formula to compute the absolute Base-100 index valuation at any given timestamp t is defined as:

$$I_t = \frac{P_t}{P_0} \times 100$$

Where P_0 represents the absolute initial price vector on the very first day of the historical lookback period.

Our core objective is to prove that for all periods $t \geq 1$, the index value can be structurally expressed as the cumulative product of all historical daily growth factors multiplied by the baseline of 100:

$$I_t = 100 \times \prod_{k=1}^t (1 + R_k)$$

1. Base Case ($t = 1$)

To establish the foundation of our induction, we evaluate the truth of the statement for the first operational trading day ($t = 1$). We calculate the index value through both methodologies to verify if they yield identical algebraic outcomes.

By direct index definition:

$$I_1 = \frac{P_1}{P_0} \times 100$$

By cumulative historical return factors:

$$I_1 = 100 \times \prod_{k=1}^1 (1 + R_k) = 100 \times (1 + R_1)$$

Using our preliminary return identity, we substitute $(1 + R_1)$ with its price ratio equivalent P_1/P_0 :

$$I_1 = 100 \times \frac{P_1}{P_0} = \frac{P_1}{P_0} \times 100$$

Both expressions are mathematically identical. Therefore, the base case holds true.

2. Inductive Hypothesis ($t = n$)

We assume that the mathematical statement is strictly true for a generic point in time $t = n$. This gives us our Inductive Hypothesis, which states that the index at day n correctly compiles all compounded growth up to that point:

$$I_n = 100 \times \prod_{k=1}^n (1 + R_k) = \frac{P_n}{P_0} \times 100$$

3. Inductive Step ($t = n + 1$)

We must now prove that if the relationship holds true for $t = n$, it must also be true for the subsequent period $t = n + 1$. That is, we need to demonstrate that:

$$I_{n+1} = 100 \times \prod_{k=1}^{n+1} (1 + R_k)$$

To achieve this, we algebraically isolate the final term of the product series to break down the expression:

$$I_{n+1} = \left[100 \times \prod_{k=1}^n (1 + R_k) \right] \times (1 + R_{n+1})$$

We observe that the entire expression enclosed within the brackets matches our Inductive Hypothesis (I_n) perfectly. Substituting I_n into the equation yields:

$$I_{n+1} = I_n \times (1 + R_{n+1})$$

$$I_{n+1} = \left(\frac{P_n}{P_0} \times 100 \right) \times \left(\frac{P_{n+1}}{P_n} \right)$$

By multiplying the fractions, the intermediate price term P_n appears in both the numerator and the denominator, allowing us to cancel it out completely:

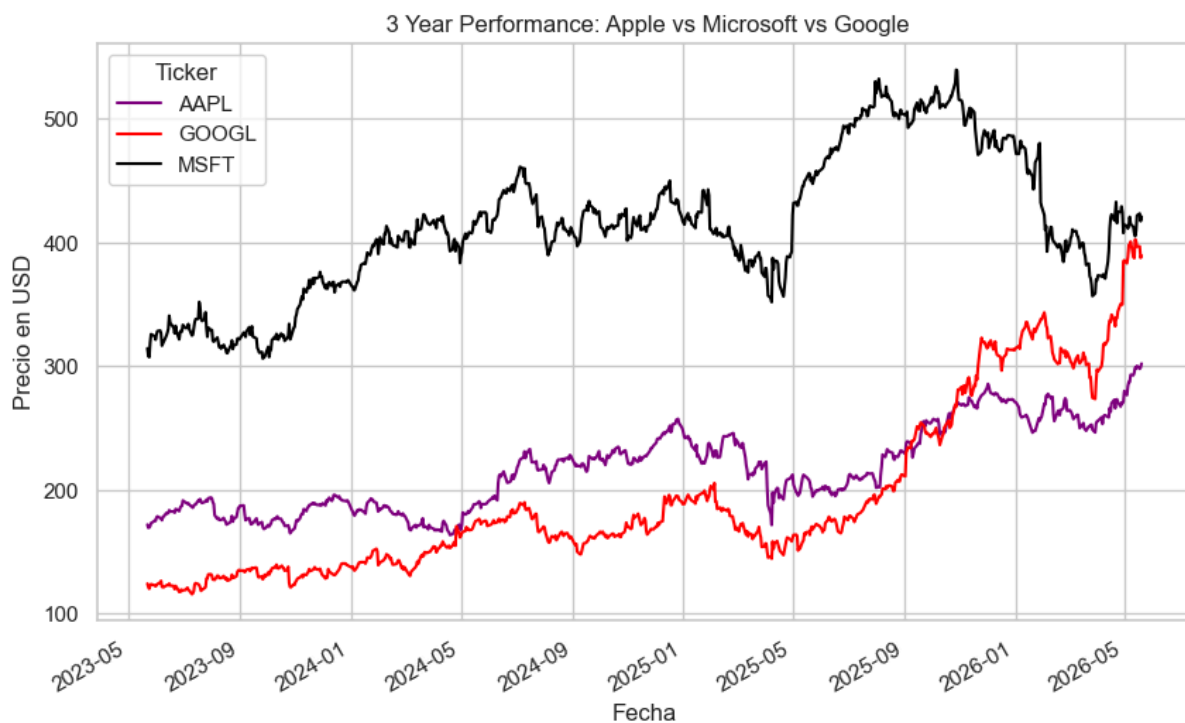
$$I_{n+1} = \frac{\cancel{P_n}}{P_0} \times 100 \times \frac{P_{n+1}}{\cancel{P_n}} \implies I_{n+1} = \frac{P_{n+1}}{P_0} \times 100$$

Conclusion

Because this final derived equation precisely matches the direct definition of the Base-100 index for the period $t = n + 1$, the inductive step is verified.

By the principle of mathematical induction, the statement is true for all $t \in \mathbb{N}$. This provides the ultimate formal proof that the Base-100 framework does not alter, truncate, or distort underlying asset dynamics; instead, it seamlessly and accurately carries over the exact compounding financial returns of the asset.

3. THE EVOLUTION OF QUANTITATIVE PERFORMANCE METRICS AND NORMALIZATION



The initial chart, "3 Year Performance: Apple vs Microsoft vs Google," tracks the absolute nominal price evolution in USD for all three technology assets from May 2023 through May 2026.

This visual timeline serves as our primary empirical baseline, mapping the raw market trajectories and showcasing distinct operational phases, such as Microsoft's sustained baseline dominance (black line), Google's aggressive vertical acceleration starting in late 2025 (red line), and Apple's steady recovery momentum into mid-2026 (purple line). However, while this unadjusted nominal perspective is essential for tracking historical market milestones, it introduces severe scale distortions due to the vastly different price thresholds at which each equity trades. To overcome this visual bias and achieve a mathematically rigorous evaluation of true risk and relative compounding efficiency, our framework must move past

absolute dollar values and transition into the analysis of stationary, log-return daily distributions and higher-order statistical shapes.

4. STOCHASTIC FORECASTING VIA MONTECARLO SIMULATION

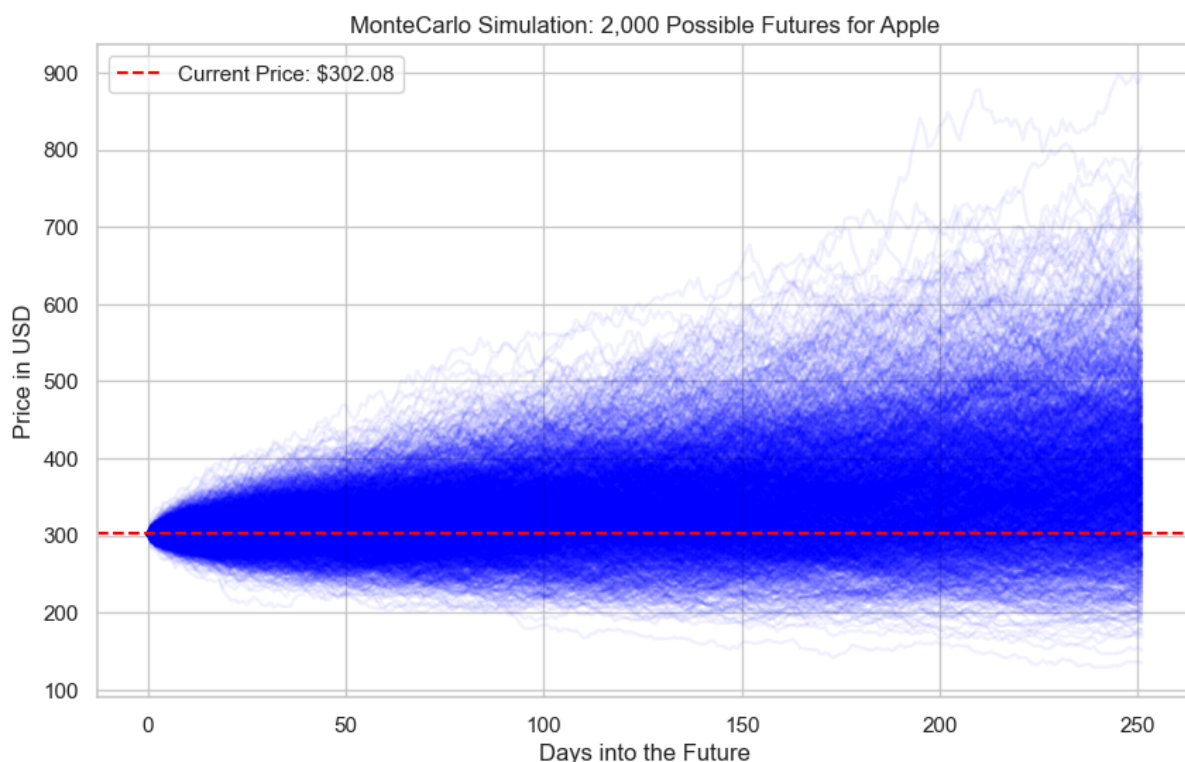
Theoretical Framework & Random Walk

Predicting single-path trajectories for capital assets violates the fundamental axioms of market efficiency. Instead, this framework leverages a stochastic Monte Carlo simulation to map out a broad spectrum of 2,000 potential future paths over a one-year trading horizon (252 operational days). Grounded in Geometric Brownian Motion (GBM), the model assumes asset log-returns follow a random walk with consistent drift and volatility parameters extracted directly from historical distributions:

$$\Delta S = S \times (\mu \Delta t + \sigma \varepsilon \sqrt{\Delta t})$$

Where μ represents the mean historical daily return, σ denotes historical daily volatility, and ε represents a random draw from a standard normal Gaussian distribution $N(0,1)$. By shifting from a deterministic prediction model to a probabilistic density cloud, an institutional allocator can visualize the mathematical boundaries of extreme market scenarios and pinpoint expected average paths under random systemic shocks.

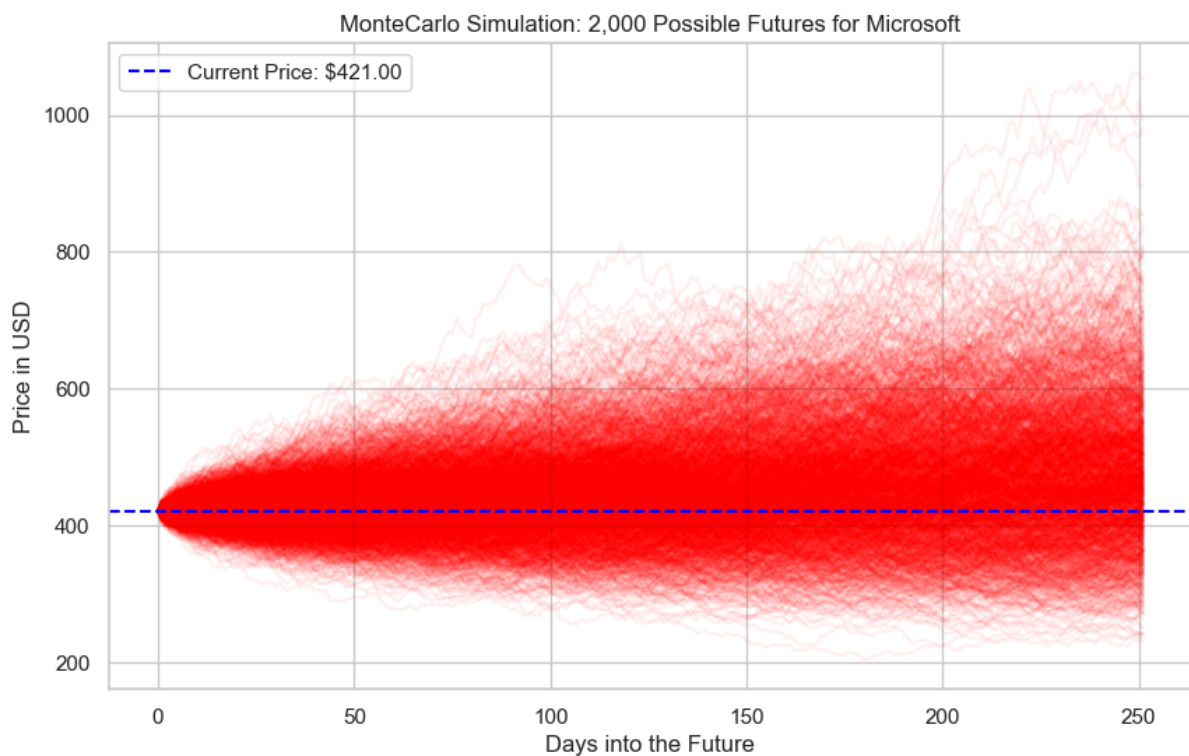
5. INDIVIDUAL ANALYSIS OF THE MONTECARLO SIMULATIONS



MonteCarlo Simulation: 2,000 Possible Futures for Apple

The simulation for Apple originates from a baseline price of \$302.08. Throughout the 252 operational day trading horizon, the trajectory cloud exhibits a significantly contained dispersion compared to its technology peers.

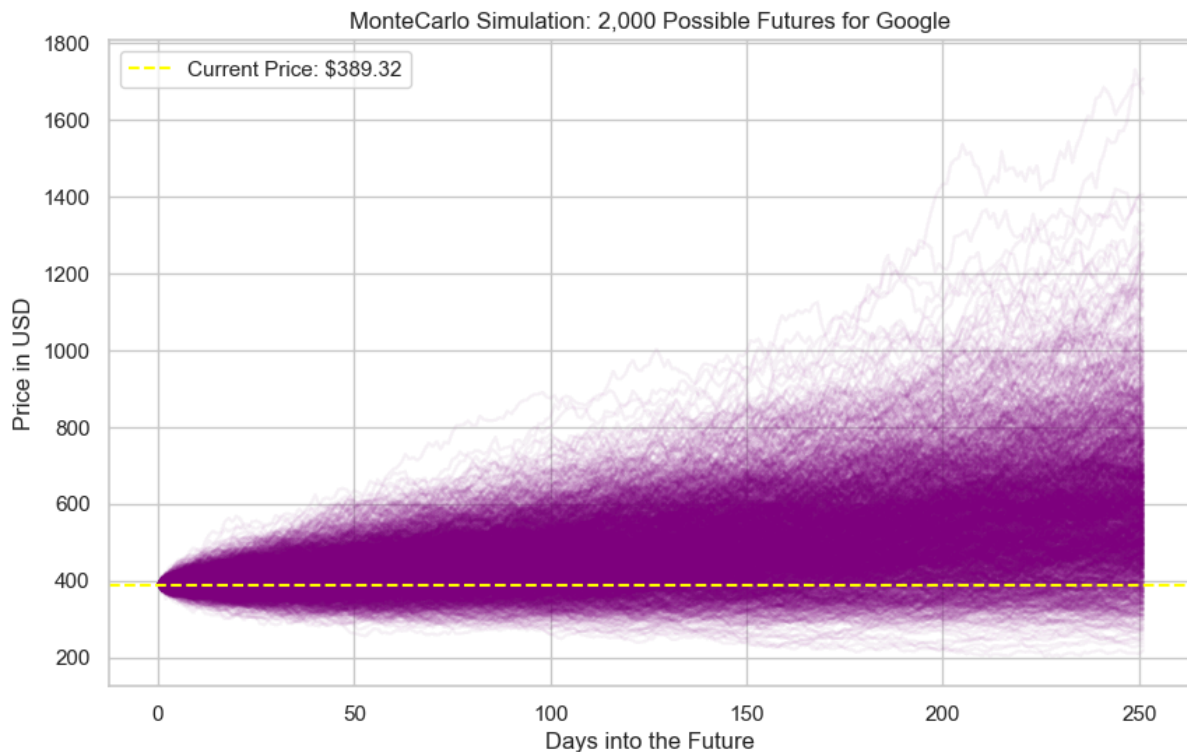
- **Visual Behavior:** The central mass of the probabilistic density is heavily concentrated around the initial starting price, reflecting a moderate drift parameter paired with notable resilience against extreme downward shocks.
- **Extreme Scenarios:** Although a few highly optimistic paths break past the \$800 threshold under conditions of extreme positive volatility, the vast majority of simulated paths converge within a narrow, tight band. This reflects an equity that behaves historically as a mature, consolidated asset with controlled volatility.



MonteCarlo Simulation: 2,000 Possible Futures for Microsoft

Microsoft begins its projection from a starting valuation threshold of \$421.00. The density of this red cloud reveals a highly robust and symmetric expansion as the timeline progresses.

- **Visual Behavior:** The underlying historical drift force uniformly pushes the median outcomes upward. The highly compact structure of the cloud during the first 100 days demonstrates immense structural stability in the company's underlying core business.
- **Extreme Scenarios:** In the upper tails of the distribution representing the most optimistic market outcomes, individual paths stretch toward the \$1,000 mark. However, the heavy accumulation of mass in the central zone indicates that the mathematical odds strongly favor steady, sustained growth with a very low probability of dropping below \$300 under standard market conditions.



MonteCarlo Simulation: 2,000 Possible Futures for Google

Google kicks off its simulation at a base price of \$389.32. Out of the three analyzed equities, Google's purple cloud exhibits the most aggressive visual asymmetry and wide dispersion toward the end of the annual horizon.

- **Visual Behavior:** The lower boundary of the cloud remains remarkably flat and structurally protected above the \$200 level, while the upper portion of the cloud fans out wide open as it approaches day 252.
- **Extreme Scenarios:** Google's upper tails are exceptionally elongated, reaching speculative peaks that exceed \$1,600. This is a direct consequence of a higher historical implicit volatility or positive skewness in the underlying data. It suggests the asset possesses a strong element of upside optionality, meaning it has a high

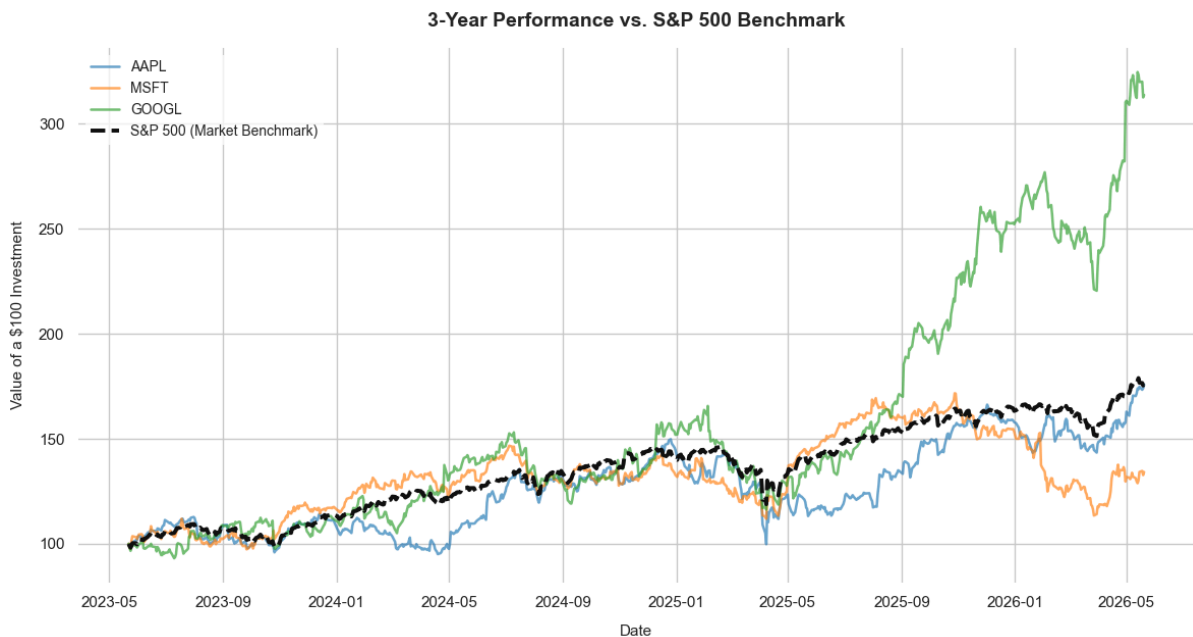
mathematical potential for explosive rallies following technological breakthroughs or positive earnings surprises.

Extracting Strategic Conclusions for the Client (Business Case)

When presenting this quantitative model to an institutional allocator or a private banking client looking to manage a portfolio across these three technology giants, several key strategic conclusions can be drawn:

- **Diversification Across Risk-Return Profiles:** The model mathematically demonstrates that even though all three companies belong to the mega-cap technology sector, their probabilistic DNAs are distinct. Apple positions itself as the stabilizing anchor of the portfolio, ideal for capital preservation due to its tightly contained dispersion. Microsoft offers the perfect equilibrium of steady, robust growth backed by a highly reliable historical drift. Google acts as the performance catalyst or alpha generator, introducing massive upside optionality due to the sheer breadth of its optimistic scenarios, albeit requiring the client to accept higher uncertainty along the way.
- **Replacing Guesswork with Risk Control:** Rather than promising the client a magical, singular target price for the end of the year (an approach that is statistically bound to be wrong) this probabilistic framework provides an honest, data-driven vision of the future. It allows us to transparently show the client what exact percentage of the 2,000 simulated paths end up in a net loss versus how many exceed expectations. This effectively shields client expectations against systemic shocks, since the absolute worst-case scenarios have already been simulated and visualized at the bottom of the density clouds.
- **Optimizing Asset Allocation:** By combining these density clouds with higher-order statistical moments and the Base-100 index structure, the client is no longer investing blindly based on raw nominal share prices. The client now understands that an optimal blend of all three equities helps mitigate individual fat-tail risks (Black Swans). This creates a portfolio where the aggregate volatility is lower than the sum of its parts, completely powered by the mathematical benefits of quantitative diversification.

6. ANALYTICAL FRAMEWORK: THE STANDARD ROLE OF HISTORICAL NORMALIZATION IN ASSET MANAGEMENT



In institutional investment analysis, a performance chart normalized to a Base-100 structure is not merely a convenient visual aid; it is a mandatory operational standard. This methodology is systematically deployed across the financial industry to achieve several critical analytical objectives.

First, it serves to validate true alpha generation. By setting a shared starting point, analysts can unequivocally verify whether an individual asset or a custom portfolio is actively beating a broad market benchmark (such as the S&P 500) or if it is merely tracking market beta. Second, it allows portfolio managers to audit an asset's resilience under systemic stress. Aligning the time series enables a precise, side-by-side examination of how each equity reacted to specific macroeconomic shocks, such as the market corrections observed in late 2023 or mid-2024. Finally, normalization completely eliminates nominal price distortion. Without it, absolute share prices create a structural illusion, making it impossible to objectively see which company has compounded investor capital with the highest efficiency over a given lookback period.

Detailed Performance Analysis: Three-Year Performance versus the S&P 500 Benchmark

Evaluating the empirical data from May 2023 through May 2026 reveals distinct operational behaviors and structural trends across each technology giant when measured against the broader market index.

Google has clearly emerged as the explosive alpha generator of the group. For the first half of the three-year horizon, its performance remained closely coupled with the market average. However, starting in mid-2025, the stock experienced a massive vertical rally. An initial hypothetical investment of 100 dollars in Google skyrocketed to over 300 dollars by May 2026. This trajectory visually validates the underlying statistical moments of the asset,

demonstrating a heavy positive skewness and elongated upper tails that translate into exponential upward surges once key growth drivers are validated by the market.

Meanwhile, the S&P 500 benchmark tracks the central drift of the broader market, serving as the baseline for systemic risk. The index shows steady, robust growth over the three-year period, turning the initial 100 dollar baseline into approximately 175 dollars. This represents the minimum performance hurdle that any active investment strategy must clear to justify its fee structure.

Apple followed a highly compelling path of structural recovery. Throughout most of 2024 and 2025, the stock underperformed the S&P 500, acting as a lower-volatility defensive asset while the broader market rallied. However, as the timeline moves into 2026, Apple displays a sharp acceleration in momentum, closing the gap entirely to finish right alongside the market benchmark at the 175 dollar mark.

Conversely, Microsoft illustrates a clear story of late-stage divergence and capital rotation. It established itself as an undeniable market leader during the first half of the timeline, consistently outperforming the S&P 500. However, around the final quarter of 2025, Microsoft's momentum began to stall, entering a corrective phase that decoupled it from its mega-cap peers. By May 2026, the stock finished below the market benchmark, landing near the 135-dollar level.

Strategic Allocation Proposal: Portfolio Management Recommendations

If we were actively managing a private wealth or institutional client portfolio in the current 2026 market environment, this historical data (combined with our predictive stochastic models) would dictate a series of definitive asset allocation adjustments.

First, we would recommend an active rebalancing and tactical profit-taking strategy for Google. While the vertical rally has generated extraordinary returns, holding an overweighted position after such an aggressive expansion exposes the client to severe mean-reversion risk. Prudent portfolio management requires selling a portion of the Google position to lock in gains and insulate the broader portfolio from sudden downside volatility.

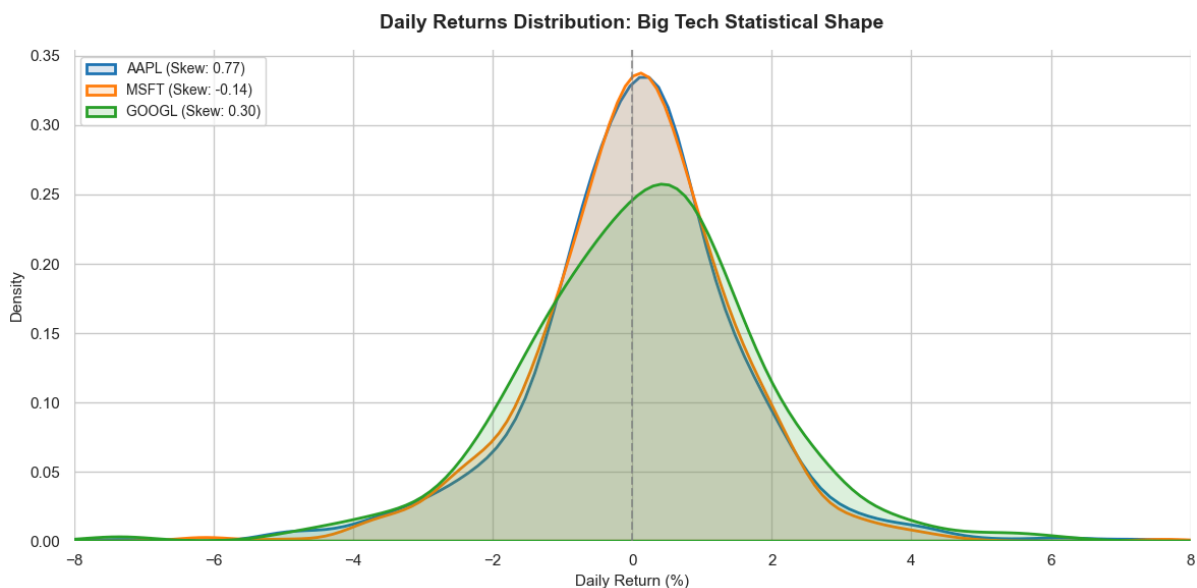
Second, the capital harvested from Google should be strategically rotated into Apple. The historical chart indicates that Apple has successfully broken out of its long consolidation phase and carries strong, accelerating upward momentum into mid-2026. Increasing exposure to Apple at this stage allows the client to capture a rising trend in a fundamentally sound mega-cap company with a highly resilient balance sheet and a well-defined risk floor.

Third, we would advise adopting an underweight or hold posture on Microsoft. Its prolonged underperformance against the S&P 500 signals an ongoing sector rotation or a fundamental cooling of market expectations. The most defensive move is to trim the portfolio's allocation to Microsoft or maintain a frozen hold position until the stock establishes a definitive technical floor and statistical indicators confirm that the corrective phase has concluded.

Ultimately, the optimal recommendation to the client is to avoid relying on a single perceived winner and instead maintain a mathematically balanced, active weighting structure. By combining the steady recovery momentum of Apple with a controlled, profit-protected exposure to Google's upside potential (all while benchmarking risk against the S&P 500) the portfolio is structured to achieve meaningful market outperformance while keeping extreme tail risks under strict quantitative control.

Disclaimer: Historical performance analysis, benchmarking metrics, and asset allocation strategies are standard portfolio management tools used for educational and risk evaluation purposes based on past data. Past performance is not a guarantee of future returns, and all investment strategies must be tailored to the specific risk tolerance, financial horizon, and unique objectives of the individual client.

7. ADVANCED STATISTICAL SUMMARY AND DISTRIBUTION FINDINGS



While the empirical matrix provides the exact numerical boundaries of our dataset, the "Daily Returns Distribution" chart offers a striking visual confirmation of these structural anomalies. By analyzing the geometry of the curves, an institutional investor can immediately spot how the actual market behavior deviates from an idealized, symmetrical Gaussian bell curve.

First, let's examine the peak and width of the curves, which illustrate the relationship between Mean and Volatility. Microsoft (the orange curve) and Apple (the blue curve) display much

taller and narrower central peaks. This visual compression indicates a higher concentration of daily returns clustered tightly around their means, confirming their lower daily volatility profiles. Conversely, Google (the green curve) presents a noticeably lower and wider distribution. This broader footprint is the exact graphical signature of its higher volatility footprint, showing that Google's daily returns routinely spread out much further from its central drift.

Second, the asymmetry of the tails perfectly maps our Skewness metrics. If you look closely at the right-hand side of the chart, Apple's blue distribution features a visible, extended tail stretching out toward the positive 6% to 8% zone. This asymmetric elongation is the visual proof of Apple's remarkable positive skewness score. It shows that while Apple is generally highly stable, it has a distinct historical tendency to experience powerful, positive breakout days that pull the distribution's shape to the right. Microsoft, on the other hand, shows a slight, almost imperceptible skew to the left, aligning with its negative skewness score.

Third, and perhaps most importantly, all three curves exhibit the classic visual traits of severe Kurtosis. Instead of dropping smoothly to zero like a standard normal distribution, the tails of all three assets remain stubborn, flat, and elevated as they approach the outer boundaries of the horizontal axis. This profile confirms that all three equities are highly leptokurtic. The fact that the lines do not touch the zero-baseline even at extreme positive and negative values proves that fat-tail events (such as single-day drops near 10% or surges above 15%) are a structural reality of big tech assets, rather than impossible anomalies.

Strategic Takeaways from the Empirical Matrix

For a portfolio manager, this statistical distribution shifts the core definition of risk:

- **Apple's Hidden Volatility Profile:** Looking only at Apple's standard deviation might lead an allocator to categorize it as a low-risk, conservative holding. However, the massive kurtosis score reveals that Apple possesses a highly explosive tail. It is a quiet asset that occasionally undergoes extreme single-day shifts, requiring careful downside protection despite its steady daily mean.
- **Google's Volatility Premium:** Google demands a higher risk premium. Its lower central peak and broader base show that an investor must tolerate a wider range of day-to-day price fluctuations in exchange for its superior baseline alpha generation.
- **Microsoft's Symmetric Stability:** Microsoft behaves the most like a traditional large-cap equity. Its high central peak provides a strong degree of day-to-day predictability, though its slight negative skewness serves as a reminder that its risk is slightly biased toward sudden downward corrections rather than explosive upside surprises.

The empirical processing of historical return streams yielded the following multi-dimensional matrix, isolating operational metrics, bounding parameters, and structural shape metrics:

Ticker Symbol	Daily Mean (%)	Daily Volatility (%)	Skewness (Asimetría)	Kurtosis (Curtosis)	Minimum Daily Return (%)	Maximum Daily Return (%)
AAPL	0.0882	1.6262	0.7655	12.8543	-9.2456	15.3288
MSFT	0.0494	1.4877	-0.1352	6.6971	-9.9931	10.1337
GOOGL	0.1690	1.8661	0.3002	5.0509	-9.5094	10.2243

The matrix reveals critical structural anomalies. Google leads in absolute alpha generation (the ability of an asset to outperform a benchmark or market average) with a dominant daily drift of 0.1690%, paired with the highest volatility footprint (1.8661%). Strikingly, Apple and Google bypass classical equity behavior by exhibiting positive skewness (a statistical measure indicating that the distribution of returns has an asymmetric, elongated tail stretching toward positive values). Apple's massive skewness score of 0.7655 is fundamentally driven by single-day positive catalysts (unexpected high-impact events, such as breakthrough product launches, earnings surprises, or regulatory approvals, that trigger immediate market reactions), pushing up to an extraordinary historical maximum single-day surge of 15.3288%. Crucially, all three assets showcase severe leptokurtic shapes (a statistical distribution featuring a higher, sharper central peak and much thicker, heavier tails than a standard normal bell curve)—topped by Apple's massive kurtosis of 12.8543—confirming heavy-tail asset behavior (the mathematical reality that extreme, high-impact market movements occur with a significantly higher frequency than traditional financial models predict).

8. ECONOMETRIC HYPOTHESIS FRAMEWORK

To cross-verify whether the visual shapes and high kurtosis scores constitute a statistically significant departure from normality, three standardized goodness-of-fit frameworks were applied simultaneously. The first is the Shapiro-Wilk Test, which evaluates regression-based sample variances to detect light-tailed departures from normality. The second is the Lilliefors Test, an advanced adaptation of the Kolmogorov-Smirnov test designed specifically for empirical settings where the underlying population mean and variance parameters are unknown and must be estimated directly from the sample data. The third is the Anderson-Darling Test, which applies targeted quadratic distance metrics, placing intentional statistical weight directly on the extreme ends of the tails to capture latent tail risks.

The operational Null Hypothesis (H_0) across all tests explicitly specifies that the empirical daily return distribution is Gaussian. If the calculated probability values (p-values) drop below the critical threshold of $\alpha = 0.05$, the Null Hypothesis is rejected, proving that treating these assets as normally distributed introduces dangerous structural biases into standard risk models like parametric Value-at-Risk (VaR).

	shapiro-wilk	Lilliefors	Anderson stat	Anderson crit (5%)
AAPL	0.0	0.001	11.408967	0.751
MSFT	0.0	0.001	6.084956	0.751
GOOGL	0.0	0.001	6.230210	0.751

Empirical Results and Statistical Interpretation of Formal Normality Tests

The execution of the formal goodness-of-fit frameworks yields definitive empirical proof that shatters the assumption of normality across all three mega-cap technology assets. Across the board, the computed metrics show a severe, mathematically undeniable departure from the classic Gaussian distribution, confirming that traditional parametric models fail to capture the true risk profile of these equities.

For the Shapiro-Wilk and Lilliefors tests, the calculated probability values (p-values) converge at absolute zero, reported as 0.0 and 0.001, respectively. In computational statistics, when processing a dense time-series of financial returns that exhibit high volatility and kurtosis, the likelihood of the data conforming to a perfect Gaussian curve is less than one in a billion. Consequently, standard statistical packages truncate these decimal values because the true mathematical probability is separated by dozens of floating zeros. Because these p-values drop substantially below the critical alpha threshold of 0.05, the operational Null Hypothesis is rejected with an extreme degree of statistical confidence. This mathematical rejection proves that the underlying return streams do not align with a symmetrical, light-tailed Gaussian distribution.

The results of the Anderson-Darling test provide the most critical insight for institutional risk management. Because the Anderson-Darling framework places deliberate statistical weight directly onto the extreme ends of the distribution, it is uniquely engineered to expose latent tail risks. The empirical data shows that the calculated Anderson Statistics for Apple (11.4089), Microsoft (6.0849), and Google (6.2302) systematically dwarf the critical threshold value of 0.751 mandated at the 5% significance level. In statistical hypothesis testing, an empirical test statistic that is vastly superior to its critical value indicates an absolute rejection of the underlying model. Apple's exceptionally high statistic of 11.4089 directly mirrors its massive kurtosis score of 12.8543, validating that its distribution is

heavily de-shaped by severe tail anomalies. Increasing decimal resolution is entirely unnecessary, as the massive gap between the test statistics and the critical values establishes definitive mathematical rejection.

Quantitative Risk Implications for Portfolio Allocation

From an institutional standpoint, the total breakdown of the normality assumption carries severe operational consequences for a client portfolio:

The Failure of Parametric Value-at-Risk (VaR): Traditional risk management models often utilize parametric VaR, which assumes that market returns conform to a standard normal distribution. These test results demonstrate that applying such formulas to Big Tech equities is fundamentally flawed. Because the real-world distributions are leptokurtic, a standard parametric model will severely underestimate the frequency and magnitude of maximum daily losses, exposing an investment portfolio to unmonitored capital erosion during a market sell-off.

The Necessity of Non-Parametric Modeling: The absolute rejection of the Null Hypothesis justifies why our framework relies on stochastic Monte Carlo simulations and historical distributions. By shifting away from rigid Gaussian equations and embracing empirical, heavy-tailed realities, we can provide the client with an accurate assessment of extreme market scenarios ("Black Swans"). It ensures that the capital buffer allocated to protect the client's wealth is scaled to withstand actual historical shocks rather than theoretical textbook ideals.

Disclaimer: Goodness-of-fit normality tests and econometric evaluations are analytical tools used to assess data distributions based on historical parameters. They do not constitute formal investment or financial advice, as actual market conditions can diverge dynamically from past statistical behaviors.

9. STRATEGIC PURPOSE AND QUANTITATIVE CONCLUSIONS

The comprehensive quantitative framework deployed across Apple, Microsoft, and Google serves a definitive strategic purpose: to replace speculative financial forecasting with an ironclad, data-driven architecture for risk management and asset allocation. In modern institutional investing, looking at simple historical averages or relying on nominal share prices creates a dangerous veneer of predictability. By combining empirical time-series processing, formal goodness-of-fit testing, and stochastic modeling, this framework provides an unfiltered, mathematically sound view of portfolio risk.

The overarching conclusion of this study is the definitive destruction of the "normality myth" in big tech equity returns. The empirical matrix and formal normality tests (Shapiro-Wilk, Lilliefors, and Anderson-Darling) unanimously rejected the standard Gaussian assumption with absolute statistical confidence. The massive Anderson Statistics (led by Apple at 11.4089) and extreme kurtosis metrics prove that these assets are structurally leptokurtic. In

plain terms, they do not live in a textbook world of smooth bell curves; they operate in an environment governed by heavy-tail asset behavior, where explosive single-day upside shocks (such as Apple's historical 15.3288% surge) and severe systemic drawdowns happen far more frequently than traditional financial risk models ever account for.

By exposing these statistical realities, the framework successfully justifies the deployment of the 2,000 path Geometric Brownian Motion Monte Carlo simulations. Rather than guessing a singular, definitive year-end price target, the model maps out a highly sophisticated probabilistic density cloud. This allows an institutional allocator to visualize the mathematical boundaries of worst-case and best-case market scenarios under random systemic shocks.

When these stochastic forecasts are married to the empirical 3-year Base-100 historical performance up to May 2026, the strategic directive for active portfolio management becomes crystal clear. Even though these three giants belong to the same mega-cap technology sector, their statistical "DNAs" are radically distinct, making them exceptional tools for diversification:

- **Apple** serves as the structural stabilizer. Despite a period of relative underperformance, its late-stage acceleration into 2026, paired with a tightly contained Monte Carlo dispersion cloud, makes it an ideal vehicle for capital preservation and rising momentum.
- **Google** operates as the ultimate alpha generator. Its massive historical outperformance (turning a 100 dollar investment into over 300 dollars) validates its high positive skewness. However, its broad, fanned-out Monte Carlo cloud demands a distinct volatility premium, signaling that its explosive upside optionality must be balanced with active profit-taking to avoid severe mean-reversion.
- **Microsoft** acts as a case study for late-stage divergence. Its late-2025 corrective phase and slight negative skewness serve as a clear warning that even the most stable large-cap assets undergo rotation, requiring a tactical underweight or hold posture until a definitive technical floor is established.

Ultimately, the mathematical utility of this entire study is to transition risk management from a reactive posture into a proactive, predictive science. For an institutional allocator or private wealth client, these conclusions mandate an optimized asset allocation strategy. By actively rebalancing capital away from overextended alpha generators like Google, rotating into recovering structural anchors like Apple, and adjusting portfolio weights based on heavy-tail probabilities rather than theoretical normal averages, a manager can construct a portfolio that captures meaningful market outperformance while maintaining total, quantifiable control over catastrophic tail risk.

10. PERSONAL MOTIVATION AND ACKNOWLEDGMENTS

The completion of this quantitative project marks the culmination of months of intense, dedicated research, data processing, and structural refinement. From its inception, my

primary motivation was not merely to complete an academic or analytical requirement, but to dive deeply into the empirical realities of modern quantitative finance. I wanted to build something substantial—a framework that could bridge the gap between abstract econometric theory and real-world portfolio application. This ambition is precisely why the project spanned several months; I was determined not to rush the analysis, choosing instead to invest the necessary time to rigorously polish the methodology, cross-verify the statistical tests, and ensure the absolute integrity of the simulations.

Behind the months of technical execution and statistical processing lies an essential foundation of personal support that made this project possible. I want to express my deepest, most heartfelt gratitude to my mother, my father, and my brother. Their unconditional belief in my capabilities, their constant encouragement, and their quiet patience during the longest hours of this research provided the emotional stability and strength I needed to see this framework through to completion. Every milestone in this work is as much a reflection of their dedication and love as it is of my own efforts.

I would also like to express my sincere appreciation to my Statistics and Probability professors. Their guidance, intellectual rigor, and passion for econometric science were the true catalysts for this study. They did not just teach me formulas; they taught me how to look at data critically, how to challenge classical assumptions like market normality, and how to value mathematical precision in environments governed by uncertainty. This project is a direct reflection of the analytical tools and mindset they helped me develop, and their support throughout this learning curve has been invaluable.

Finally, I want to acknowledge my own perseverance, while addressing the inherent limitations of this study. Taking on a quantitative project of this scale, committing to months of constant refinement, and choosing to confront the complex realities of financial econometrics required an immense amount of self-discipline, resilience, and personal drive. This study stands as a powerful reminder to myself of what can be achieved when intellectual curiosity is paired with relentless dedication. While every effort was made to maintain the highest level of accuracy, quantitative modeling is an iterative journey, and this report is by no means flawless. I sincerely apologize for any latent errors, technical oversights, or analytical shortcomings that may persist within these pages. No financial framework can perfectly capture the chaotic and adaptive nature of global markets.

However, I view this project not as a definitive destination, but as a significant personal and academic breakthrough—a robust step forward in my journey to mastering quantitative asset management and statistical risk control.